

OSIPI · GSOC 2026 · ASL QUALITY CONTROL

Kidney QC Design Brief

Extending the ASL QC ToolBox from brain to kidney

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KIDNEY_QC_BRIEF.md · 2,764 words

reconciled against osipy-qc · 20 registered checks · 253 tests



The one thing to take away

The renal ASL literature has an authoritative, consensus-graded specification for **how a renal ASL scan should be acquired**, and essentially nothing on **how to judge whether the resulting perfusion map is good**. That inverts the brain toolbox, where Stream B is the rich half (QEI with a validated ≈ 0.5 cutoff, sCoV 0.67, GM/WM 40–100) and Stream A is thin. In kidney it is the other way round, and the design has to say so rather than build a symmetrical doc on numbers that do not exist.



The central finding

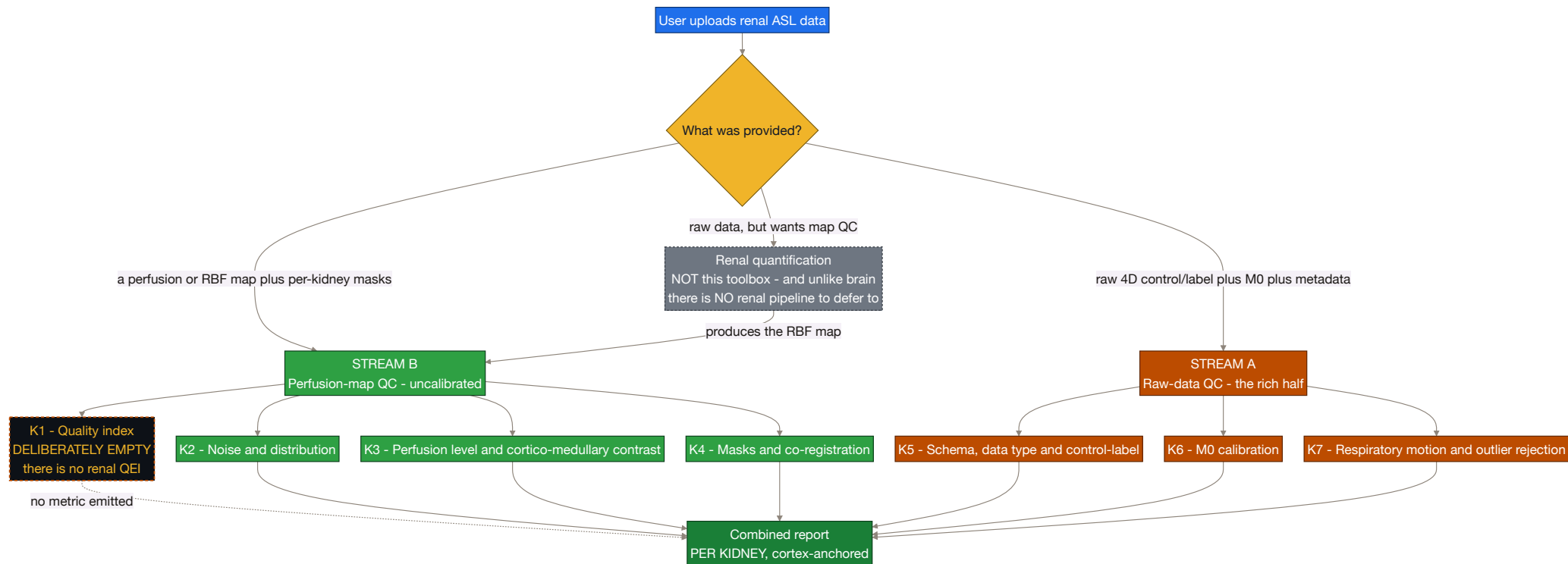
Nery et al. 2020 (MAGMA 33:141–161; PARENCHIMA COST CA16103) is the renal equivalent of the 2015 ASL White Paper: "*An international panel of 23 renal ASL experts followed a modified Delphi process*" (p.1), and "*Fifty-nine statements achieved consensus, while agreement could not be reached on two statements*" (p.1). It gives a hard Stream A protocol spec — labelling, readout, background suppression, pair counts, M0 rules, quantification constants, all with agreement percentages.

It contains zero numeric image-quality thresholds. No tSNR floor, no CoV cutoff, no motion limit, no negative-voxel fraction, no QEI analogue. Its closest approach to a quality statement is qualitative: "*This approach is considered robust and reproducible and can provide renal perfusion images of adequate quality*" (p.1). Even the QC-adjacent statements specify no method and no number — "*8.2 Outlier rejection is recommended for renal ASL*" (p.8; 0% abstentions, **100% agreement**), backed in the body only by "*Outlier rejection methods ... have relied on manual ...*" (p.12) plus a citation list.

Consequence: every renal CBF-map (Stream B) threshold in this design ships tagged  **UNCALIBRATED**, and an uncalibrated threshold may never drive a FAIL on its own.

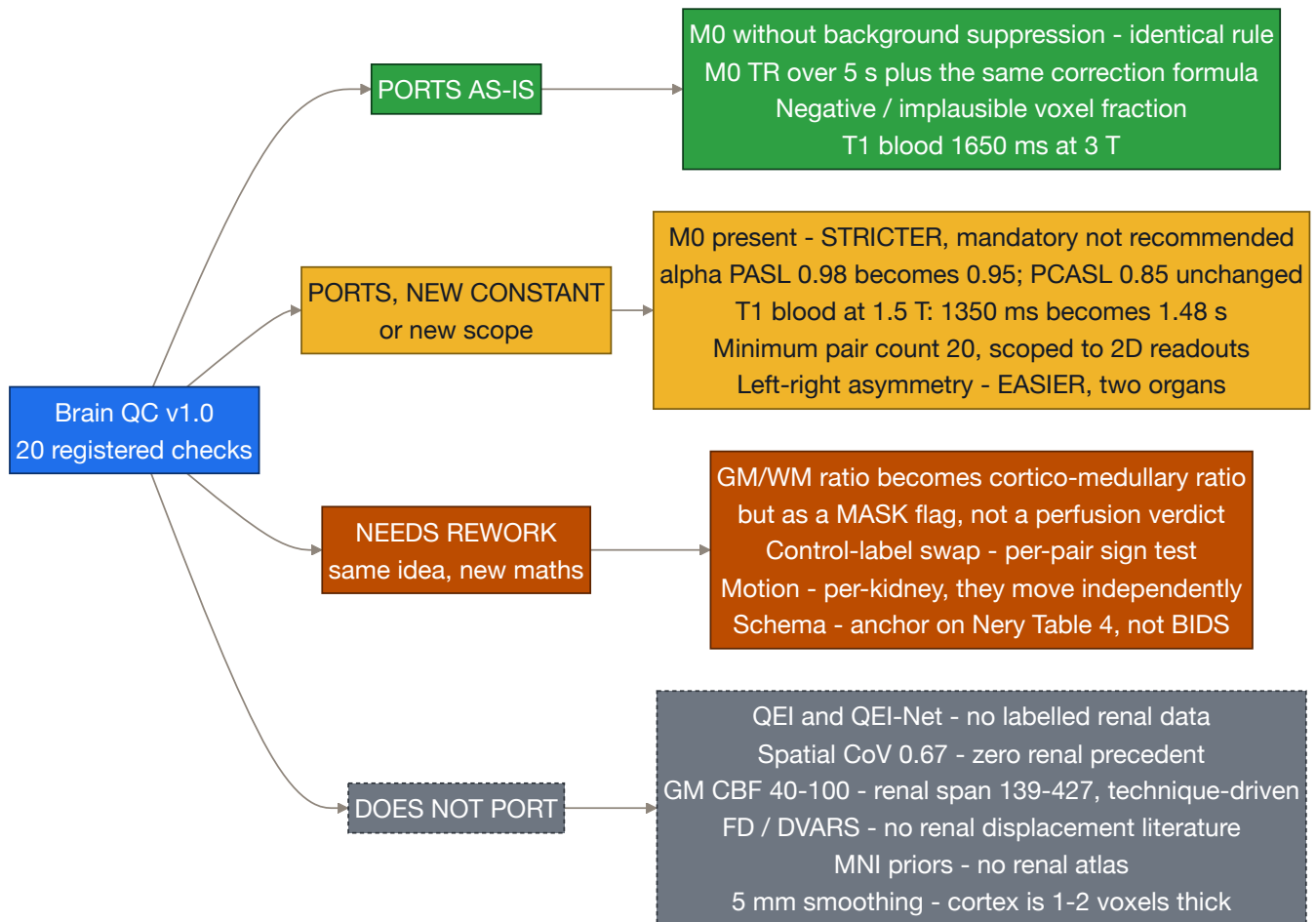


Proposed module map



K1 is drawn hollow because it is empty on purpose. The brain anchor check does not port: there is no cortex/medulla probability substrate to build a **spRBF** template from, and — the binding constraint — **no renal ASL dataset with expert quality ratings that I could find**, so there are no curves to fit and no cutoff to validate. Shipping a "renal QEI" with brain-fitted constants would be the single most misleading thing this toolbox could do. The slot stays visible so the gap is visible.

🔄 What actually transfers from brain



The number worth staring at in that bottom box: published healthy **cortical** perfusion spans **139–427 mL/100 g/min** — "Renal cortical perfusion by ASL ranged from 139 to 427 mL/100 g/min in healthy volunteers" (Odudu 2018, PARENCHIMA systematic review over 53 studies to Jan 2018, p.2, aggregated in Supplementary Table S1). The patient range, **83–412**, overlaps it almost entirely. And the dominant driver of that spread is **technique, not biology**: Harteveld 2020 measured FAIR and pCASL in the **same 16 healthy volunteers at 3 T** and got cortical RBF **362 ± 57** vs **201 ± 72 mL/min/100 g** — a **1.8× gap from the labelling scheme alone**. A brain-style "normal CBF band" is therefore not available in kidney at any useful tightness.

The proposed checks

ID	WHAT IT ASKS	TIER
K1.1	(renal quality index — not implemented in v1; no labelled data exists)	—
K2.1	Temporal SNR per kidney — report-only	 UNCALIBRATED
K2.2	Is perfusion-weighted signal a plausible % of M0? (band ~0.5–8%)	 values published,  band
K2.3	Negative and implausible-value fraction, cortex-restricted	 0–500 clip
K2.4	Cortex-restricted spatial CoV (<i>stretch</i> — no renal precedent)	 UNCALIBRATED
K3.1	Cortical RBF level, per kidney — the consensus-mandated quantity	 range,  verdict bands
K3.2	Cortico-medullary ratio — as a segmentation-integrity flag	 UNCALIBRATED (never FAILs)
K3.3	Left-vs-right cortical consistency	 UNCALIBRATED
K4.1	Per-kidney mask integrity (size, contiguity, overlap)	 DEFINITION
K4.2	ASL ↔ M0 and per-kidney registration	 UNCALIBRATED
K4.3	Slice coverage / usable-slice fraction	 DEFINITION
K5.1	Metadata completeness vs Nery Table 4 (not BIDS — BIDS has no kidney)	 PUBLISHED
K5.2	Data-type / geometry detection — routing, INFO only	 DEFINITION
K5.3	Control-vs-label ordering, per-pair sign consistency	 DEFINITION
K6.1	M0 present — " <i>M0 acquisition is mandatory</i> " (R9.1, 94%)	 PUBLISHED
K6.2	M0 without labelling and without background suppression	 PUBLISHED
K6.3	M0 TR > 5 s, else corrected	 PUBLISHED
K6.4	M0 geometry matches ASL (<i>stretch</i>)	 DEFINITION
K7.1	Per-kidney respiratory displacement	 UNCALIBRATED
K7.2	Subtraction-outlier rate, surviving-pair count	 IMPLEMENTATION ×4
K7.3	Breathing strategy and gating efficiency	 PUBLISHED
K7.4	Spurious-labelling signature (<i>stretch</i>)	 UNCALIBRATED

 = a paper states this number for this purpose ·  = a reference study's code/method uses it, not a validated cutoff ·  = engineering default, no published source ·  = maths, nothing to tune.

Two published rules that are hard and checkable: **20 ASL pairs minimum** — "*4.7 In single-T1 acquisitions, a minimum of 20 ASL pairs is recommended*" (p.7; **89%**) and its identically worded PCASL twin "*5.12 In single-PLD acquisitions...*" (p.7; **83%**); and the labelling efficiencies "*9.8 Labeling efficiency PCASL = 85%*" (p.8; **86%**) and "*9.7 Labeling efficiency PASL = 95%*" (p.8; **100%**) — renal PASL is **95%**, not the brain White Paper's 98%.

Agreement percentages gate severity. "6.4 Spin-echo EPI is the preferred readout for 2D single-slice acquisitions" sits at **75%** (p.7) — the weakest of all 59 statements, exactly on the inclusive consensus bar — and "7.4 Renal ASL scans under free breathing are preferred" at **76%** (p.8) is second weakest. Both must **WARN**, **never FAIL**. By contrast "7.3 Breath-hold scans are not recommended for clinical renal ASL" carries **94%** (p.8), so detecting a breath-hold is a firmer basis for a flag than the mere absence of free breathing.

Minimum inputs — and what a reviewer may conclude at each tier

TIER	YOU MUST SUPPLY	WHAT A REVIEWER MAY LEGITIMATELY CONCLUDE
B0	RBF map only	Almost nothing — global histogram shape and implausible-value fraction. There is no renal analogue of "just BET the brain", so a lone map buys far less than in brain.
B1	+ whole-kidney masks, L and R separately	"Nothing is grossly broken" — negative fraction, mask integrity, coverage, left-vs-right. Not the consensus quantity.
B2	+ cortex mask per kidney	The one conclusion the consensus actually asks for: cortical RBF, per kidney, in a published context.
B3	+ medulla mask (or a quantitative T1 map to derive one)	Additionally: whether the cortex and medulla masks are genuinely separated. Never a statement about the medulla's perfusion.
A0	raw 4D control/label + kidney masks	Whether the acquisition survived breathing — outlier rate, per-kidney displacement, pair ordering.
A1	+ M0 / PD image	Whether calibration is usable at all: M0 presence, ASL↔M0 registration, PWS as % of M0.
A2	+ acquisition metadata (JSON or vendor header)	Full protocol conformance vs the renal consensus — the richest and best-sourced verdict this toolbox can give.
A3	+ respiratory trace	Gating efficiency, and whether motion explains the rejected pairs.

The renal cliff is **masks, not tissue maps** — R9.10 (100%) makes manual ROI the default and R9.11 (93%) says ROIs are drawn on the M0 or a structural image, not on the perfusion map, so nothing here auto-derives the way a brain mask does. **Cortex masks are the highest-value input a user can add.**

The two rules we get for free

These are the only two places where the renal consensus hands us a QC-shaped constraint outright, and both are about **what to report**, not about quality:

R10.1 — "Cortical renal blood flow values (not whole-kidney) should be reported, separately for left and right kidney." — p.8, **0% abstentions, 100% agreement**

Unanimous, zero abstentions — the cleanest published constraint in the renal corpus, and it drives the **data model**, not just a check: the report object is `{left: {...}, right: {...}}`, the aggregator runs per kidney then combines, and no check may headline a single whole-organ number.

R10.2 — "Medullary renal blood flow values are not considered reliable with current measurement approaches."
— p.8, **14% abstentions, 89% agreement**

Independently corroborated by measurement: medullary reproducibility spans **ICC 0.27–0.94, CV 3–43%** within visits and **ICC 0.13–0.96, CV 4–37%** between visits (📄 Odudu 2018, p.3–4), against cortical **ICC 0.62–0.98 / CV 3–18%** and **ICC 0.85–0.97 / CV 4–13%**; and Garcia-Ruiz 2025 reports medullary RBF ICCs of **0.081–0.426**.
Rule: no medullary metric may drive a FAIL.

★ Why cortex:medulla ratio is a mask check, not a perfusion verdict

This is the design decision most likely to be challenged, so here is the whole argument in one place.

`CMR = mean(RBF[cortex]) / mean(RBF[medulla])`, per kidney — the renal analogue of the brain GM/WM ratio in **form** (scale-free, immune to a global quantification error) but deliberately **not in role**. In brain, GM/WM < 1 is a published data-quality FAIL. In kidney, CMR may flag the **masks** and may never grade the kidney. Three reasons:

1. **Its denominator is the least reliable measurement in renal ASL** — R10.2 above, plus ICCs down to 0.081. Grading a scan on CMR means grading it on the one number the consensus says not to trust.
2. **It is not monotone with disease.** Across the two severity-staged cohorts in the paper library it moves in *opposite* directions — rising 1.57 → 2.01 with CKD stage in one, falling 2.08 → 1.75 in the other. A quantity that moves both ways cannot discriminate disease.
3. **But it does detect one specific, common, published failure.** Only ~10% of renal blood flow perfuses the medulla (📄 Odudu 2018, p.1), so a CMR near 1 is anatomically impossible — evidence the two masks are drawing from the same tissue. Worked example: Hammon 2016 reports CMR ≈ 1.21 and the authors say so themselves — "*the segmentation of the medulla may contain parts of the cortex what explains higher than expected medullary perfusion values*". Clean studies cluster at 2.26–2.59.

So the check emits **INFO** normally, **WARN** only in the low direction with the reason "*cortex and medulla masks may not be separated*", **never FAIL**, and is excluded from aggregation. The ~1.5 trip point is 🔧
UNCALIBRATED — no paper states a CMR threshold for any purpose — and sits between the two clusters above with margin. High CMR (5–8.5) carries no verdict: it just means a strict inner-medulla ROI convention.

? Open questions — the answers change the design

For the renal specialist, above all:

1. **What rejected-pair count makes a renal scan unusable?** Four published outlier rules exist (± 2 SD / >20% of voxels, and variants) and **not one paper says how many rejected pairs is too many**. Harteveld's rule fires in ~2/3 of *healthy* datasets with a ceiling of 2 rejected pairs. Is " ≥ 3 rejected → WARN, surviving pairs <

20 → FAIL" defensible — or is applying the 20-pair *acquisition* recommendation to *surviving* pairs a misuse of it? **Is there an unpublished in-house standard the renal community actually runs?**

2. **Is there a renal heterogeneity metric worth having at all?** sCoV has zero renal precedent, and a whole-kidney CoV would be dominated by the normal cortico-medullary gradient — i.e. by anatomy, not artefact. Is cortex-restricted CoV meaningful, or is cortico-medullary CNR the better primitive?
3. **Is CMR-as-a-mask-check the right framing, or too clever?** Would a renal reader actually act on "CMR 1.18, masks may not be separated"?

For the mentors:

4. **Cortex-only, or cortex + medulla?** CMR needs a medulla mask that most users will not have. Do we (a) require it, (b) derive it from a T1 map when present, or (c) ship CMR as optional? My inclination is (b) with (c) as fallback — this decides whether the toolbox needs a segmentation step at all.
5. **Which kidney T1 for the M0 TR correction?** The consensus prescribes $SI_{PD}/(1 - \exp(-TR/T1, \text{tissue}))$ (p.13) and then never supplies a value; cortex and medulla differ by ~400 ms. Apply one default, or refuse to correct and WARN with the magnitude under both compartments (my current proposal)?
6. **Do we ever compute RBF ourselves, or only consume it?** The brain toolbox settled this as "QC layer, light derivation only" and deferred to PyASL/ASLPrep. **There is no renal pipeline to defer to** — ASLPrep, ExploreASL and PyASL contain zero renal code, `ukat` has no ASL module. Same line?
7. **Is there any renal ASL dataset with expert quality ratings, or a repeat-scan dataset?** Without labels there is no renal QEI and no calibrated Stream B threshold — the binding constraint on this entire design. Could PARENCHIMA or the ISMRM Renal MR Study Group supply one?
8. **Taso et al. 2023 (MRM 89:1754–1776), the ISMRM body-ASL "Grey Paper", is paywalled and I have not read it.** Its organ-by-organ review starts with kidneys and its senior author is Nery 2020's. Could María supply a copy — and does it supersede any Nery parameter?

Every number above carries its provenance tier. Where nobody has published one, this brief says so rather than inventing a band — the same policy as the brain design, in a domain where it bites harder.